

Lab 4

Stephanie Salgado

Setup:

Creating an Experiment

New Experiment

Project

csci4306

Creator

csci3306ogrp

Name

stephaniesalgadolab4

Valid experiment name

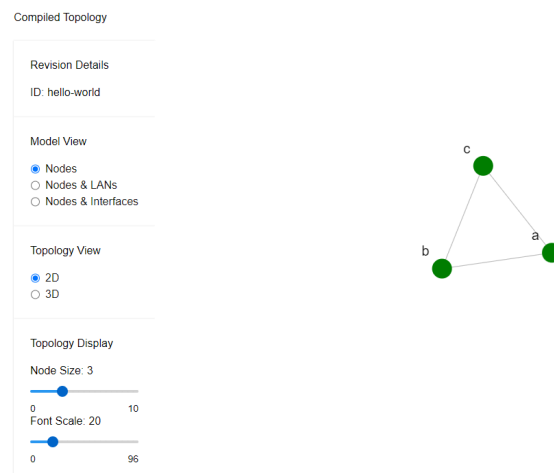
Description

Lab4

Submit

Similar to Lab 1, I created a new experiment from the “Experiments” tab.

Compiled Topology



With the same model from Lab 1, I compile it in the “Model Editor” tab.

Pushing Model to Experiment

Push Model to Experiment ×

Experiment *

stephaniesalgadolab4.csci4306

Branch

master

Tag

Push

Then, I push the model to the new experiment for Lab 4.

Realizing the Experiment

Reserve Revision ×

⚠

A Reservation is a request to exclusively reserve resources for an experiment. This is an asynchronous process. Check the reservations listing or the dashboard to see the status of your reservation. If the reservation fails, it is likely that your model is requesting resources that are not currently available. In that case, relinquish the reservation and try again at a later time.

Project

csci4306

Experiment

stephaniesalgadolab4

Reservation Name

stephaniesalgadolab4

Submit

Materializing the Realization



Success

[stephaniesalgadolab4.stephaniesalgadolab4.csci4306](#)

 Metal Nodes: 0

 Virtual Nodes: 3

 Links: 2

 Ingresses:

 Created: 9/24/2024, 12:27:40 PM

 Last Updated: 9/24/2024, 12:27:49 PM

[View Details](#)

Attaching XDC

New Experiment Development Container

Project

csci4306

Name

stephaniesalgadolab4

Valid

Type

shared

Submit

Attach to Materialization

XDC

stephaniesalgadolab4.csci4306

Materialization

stephaniesalgadolab4.stephaniesalgadolab4.csci4306

Attach

After all these steps, I successfully completed the setup for this lab.

Setting up my account

```
Terminal 1
# su - csci3306ogrp
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

csci3306ogrp@stephaniesalgadolab4:~$ echo "PATH=$PATH:/share" > .profile
csci3306ogrp@stephaniesalgadolab4:~$ echo ufxKhQaAha > pass.txt
csci3306ogrp@stephaniesalgadolab4:~$ exit
logout
# su - csci3306ogrp
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

csci3306ogrp@stephaniesalgadolab4:~$ # su - csci3306ogrp
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

csci3306ogrp@stephaniesalgadolab4:~$ echo "PATH=$PATH:/share" > .profile
csci3306ogrp@stephaniesalgadolab4:~$ echo ufxKhQaAha > pass.txt
csci3306ogrp@stephaniesalgadolab4:~$ exit
logout
# su - csci3306ogrp
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

csci3306ogrp@stephaniesalgadolab4:~$
```

I follow steps 1-5 to set up my account.

Running the lab

```
csci3306ogrp@stephaniesalgadolab4:~$ startexp twonode
Path /share/education lab twonode name twonode proj csci3306ogrp
Logging csci3306ogrp into Merge
Creating experiment twonode.csci3306ogrp
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Waiting for "mrg new experiment twonode.csci3306ogrp" to succeed. Rerunning.
Error creating new experiment.

csci3306ogrp@stephaniesalgadolab4:~$ runlab twonode
Warning: Permanently added 'a' (ED25519) to the list of known hosts.
install
Warning: Permanently added 'b' (ED25519) to the list of known hosts.

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.
```

I used “startexp twonode” to start the “twonode” experiment, then I used “runlab twonode” to run it.

Ping

```
csci3306ogrp@stephaniesalgadolab4:~$ ssh a
Linux a.infra.helloworldssalgado.stephaniesalgadolab4.csci4306 5.10.0-29-amd64 #1 SMP Debian 5.10.216-1 (2024-05-03) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Sep 24 18:38:01 2024 from 192.168.254.2
csci3306ogrp@a:~$ ping b
PING b.exp.helloworldssalgado.stephaniesalgadolab4.csci4306 (10.0.1.2) 56(84) bytes of data.
64 bytes from 10.0.1.2 (10.0.1.2): icmp_seq=1 ttl=64 time=0.412 ms
64 bytes from 10.0.1.2 (10.0.1.2): icmp_seq=2 ttl=64 time=0.351 ms
64 bytes from 10.0.1.2 (10.0.1.2): icmp_seq=3 ttl=64 time=0.368 ms
64 bytes from 10.0.1.2 (10.0.1.2): icmp_seq=4 ttl=64 time=0.388 ms
64 bytes from 10.0.1.2 (10.0.1.2): icmp_seq=5 ttl=64 time=0.355 ms
^C
--- b.exp.helloworldssalgado.stephaniesalgadolab4.csci4306 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4008ms
rtt min/avg/max/mdev = 0.351/0.374/0.412/0.022 ms
```

I used “ssh a” to log into node a, then pinged b, using “ping b”.

Iperf

```
csci3306ogrp@a:~$ iperf -s
-----
Server listening on TCP port 5001
TCP window size: 128 KByte (default)
-----
[]

csci3306ogrp@b:~$ iperf -c a
-----
Client connecting to a, TCP port 5001
TCP window size: 340 KByte (default)
-----
[ 3] local 10.0.1.2 port 43436 connected with 10.0.1.1 port 5001
[ ID] Interval      Transfer    Bandwidth
[ 3] 0.0000-10.0000 sec 23.2 GBytes 19.9 Gbits/sec
csci3306ogrp@b:~$ []

csci3306ogrp@a:~$ iperf -s
-----
Server listening on TCP port 5001
TCP window size: 128 KByte (default)
-----
[ 4] local 10.0.1.1 port 5001 connected with 10.0.1.2 port 43436
[ ID] Interval      Transfer    Bandwidth
[ 4] 0.0000-10.0007 sec 23.2 GBytes 19.9 Gbits/sec

csci3306ogrp@b:~$ iperf -c a > iperf.out
```

From node a, I use “iperf -s” which starts a iperf server. Then from node b, I used “iperf -c a”. After that, node a reported the bandwidth it measured. Last, I used “iperf -c a > iperf.out” to store the output into a file.

Copying Files

```
PS C:\Users\Stephanie\Desktop> .\mrg config set server grpc.sphere-testbed.net
PS C:\Users\Stephanie\Desktop> .\mrg login csci3306ogrp
Enter password for csci3306ogrp:
```

```
PS C:\Users\Stephanie\Desktop> .\mrg xdc scp download -x stephaniesalgadolab4.csci4306 b:iperf.out .
level=info msg="downloading b:iperf.out to . via stephaniesalgadolab4.csci4306 ..."
PS C:\Users\Stephanie\Desktop> dir

Directory: C:\Users\Stephanie\Desktop

Mode                LastWriteTime         Length Name
----                -
-a----           9/24/2024   3:13 PM           362 iperf.out
-a----           8/23/2024   8:41 AM       26226176 mrg.exe

PS C:\Users\Stephanie\Desktop>
```

I run the provided commands and notice, “iperf.out” is now on my desktop.

```
# su - csci3306ogrp
csci3306ogrp@stephaniesalgadolab4:~$ echo "Hello world" > hw.out
csci3306ogrp@stephaniesalgadolab4:~$
```

```
PS C:\Users\Stephanie\Desktop> .\mrg xdc scp download stephaniesalgadolab4.csci4306:hw.out .
level=info msg="downloading stephaniesalgadolab4.csci4306:hw.out to . ..."
PS C:\Users\Stephanie\Desktop> dir
```

Directory: C:\Users\Stephanie\Desktop

Mode	LastWriteTime	Length	Name
-a----	9/24/2024 3:19 PM	12	hw.out
-a----	9/24/2024 3:13 PM	362	iperf.out
-a----	8/23/2024 8:41 AM	26226176	mrg.exe

On the XDC, I copy “Hello world” onto a file named “hw.out”. On my machine, I’m able to download it. Notice, it’s now on my desktop like “iperf.out”.

```
PS C:\Users\Stephanie\Desktop> echo "Local file" > lf.out
PS C:\Users\Stephanie\Desktop> .\mrg xdc scp upload -x stephaniesalgadolab4.csci4306 lf.out b:.
level=info msg="uploading lf.out to b:. via stephaniesalgadolab4.csci4306 ..."
PS C:\Users\Stephanie\Desktop>
```

```
csci3306ogrp@b:~$ ls
iperf.out lf.out
csci3306ogrp@b:~$
```

This time, I want to upload and copy a file from my machine, to the experimental nodes. Notice, after uploading, the “lf.out” file is now on node b.

```
PS C:\Users\Stephanie\Desktop> echo "Local file take two" > lftt.out
PS C:\Users\Stephanie\Desktop> .\mrg xdc scp upload lftt.out stephaniesalgadolab4.csci4306:.
level=info msg="uploading lftt.out to stephaniesalgadolab4.csci4306:. ..."
PS C:\Users\Stephanie\Desktop> dir
```

Directory: C:\Users\Stephanie\Desktop

Mode	LastWriteTime	Length	Name
-a----	9/24/2024 3:19 PM	12	hw.out
-a----	9/24/2024 3:13 PM	362	iperf.out
-a----	9/24/2024 3:23 PM	26	lf.out
-a----	9/24/2024 3:27 PM	44	lftt.out
-a----	8/23/2024 8:41 AM	26226176	mrg.exe

```
csci3306ogrp@stephaniesalgadolab4:~$ ls
hw.out  lftt.out  pass.txt
csci3306ogrp@stephaniesalgadolab4:~$
```

Lastly, I copy some text onto “lftt.out” from my machine then upload it. On the XDC, I’m able to find the same file, as well as the other 2 files previously on it.

Stopping the Experiment

```
csci3306ogrp@stephaniesalgadolab4:~$ stopexp twonode
Path lab twonode name twonode proj csci3306ogrp
Configuring mrg
Logging csci3306ogrp into Merge
Checking if the experiment is realized.
Error reading twonode revision.
Deleting entire experiment.
```

Finally, on the XDC, I use “stopexp twonode” to end the experiment.

Summary

This lab followed much of the setup from Lab 1. When trying this out, I encountered some issues with step 7 (startexp twonode). However, after running step 8 (runlab twonode), everything seemed to work. I was able to successfully get mrg on my windows machine. I was able to download a file from the experimental node to my machine and vice versa. I was also able to do the same when using the XDC. Overall, I found this lab somewhat time consuming but I thought it was very interesting how I was able to download and upload files between my machine, the nodes and my XDC.